

Sleep and Sleep Disorders

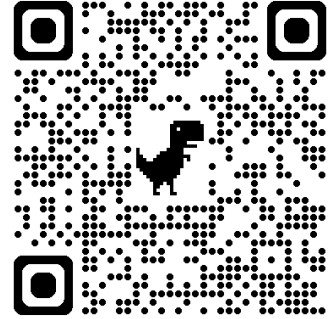
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Do.
Make.
Heal.

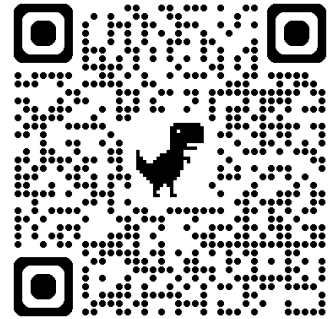
Innovate.

Reinvent the Future.

- Session Objectives:
 1. Understand the basics of sleep physiology
 2. Recognize, diagnose and manage patients with sleep disorders, including use of the polysomnogram for evaluation and when to obtain outside consultation
 3. Understand the importance of sleep wellness to optimize provider cognition and patient safety
- Connections
 - ScholarRx: Sleep physiology; Sleep Disorders: QR→
 - Lecture on Pharmacology of Sedatives/Hypnotics (PPOM1)
 - Lecture on Sleep Physiology and Sleep Apnea (PPOM2)



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Sleep Physiology

Sleep in Health & Human Development

- Recommended: 7-9 hours (h)/night
 - Americans average 6.9 h
 - Requirement varies with age
 - Newborn 16-20 h
 - Child 10-12 h
 - Mid-adolescence 9-10 h (weekend catch-up)
 - Young adult 7-7.5 h
- Pattern is also regional and changes with aging
 - Siesta
 - Earlier sleep time in older adults

Sleep Wake Cycle

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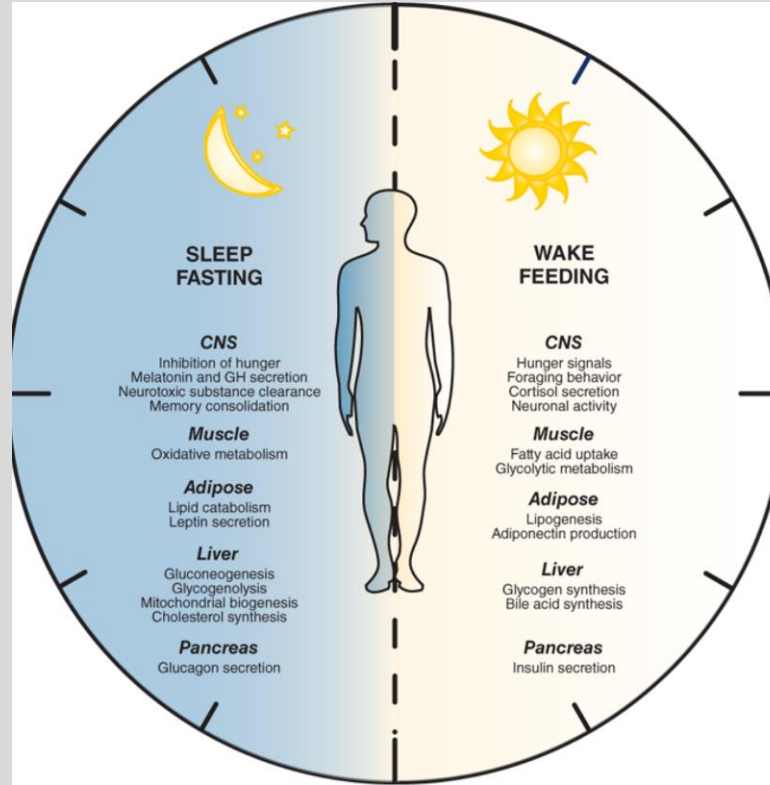
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Fun Fact 1!

Circadian rhythms are present in almost all living things.

Fun Fact 2!

Circadian rhythms result in physical, mental and behavioral changes.

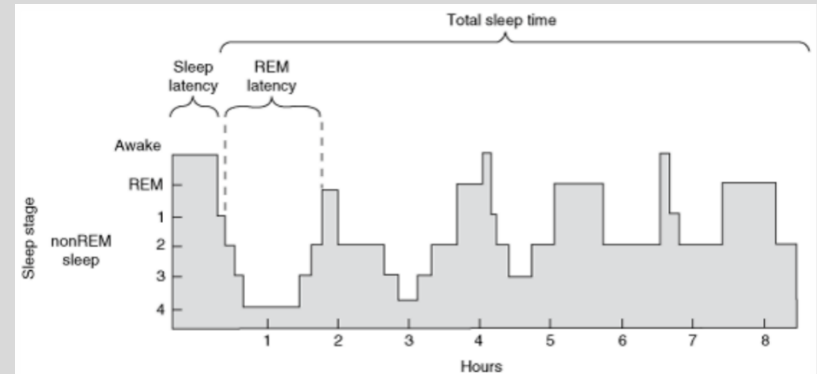
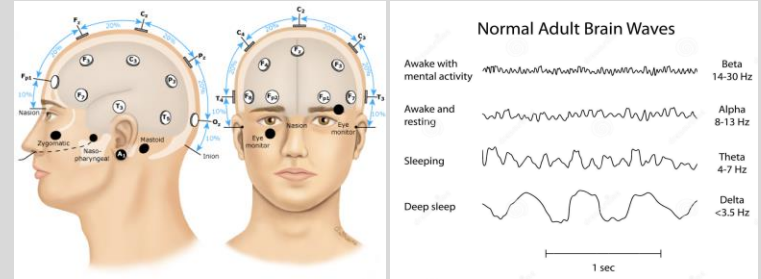


Fun Fact 3!

Our body's clock cycle without the influence of the day-night cycle is actually 25 hours.

Measuring Sleep: The Polysomnogram

- Overnight sleep study
- EEG, EOG, EMG, ECG, airflow, respiratory effort, O2 saturation, limb movements, etc.
- Usually performed in sleep lab (Neurology, Pulmonary)
- Gold standard to diagnosis sleep apnea, narcolepsy, parasomnias, RLS



Source: Ebert MH, Loosen PT, Nurcombe B, Leckman JF:
Current Diagnosis and Treatment: Psychiatry, 2nd Edition:
<http://www.accessmedicine.com>

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EEG = electroencephalogram, EOG = electrooculogram, EMG = electromyogram,
ECG = electrocardiogram, RLS = restless legs syndrome, REM = rapid eye movements

Adapted from: Jasper HH. Report of the committee on methods of clinical examination in
electroencephalography: 1957. Electroencephalogr Clin Neurophysiol 1958; 10:370.

Sleep Architecture

AASM	EEG	Other features
W	Posterior alpha 9-11 Hz intermixed low-voltage fast activity of mixed frequency	Electromyogram is silent except for facial muscles
N1	Progressively lower voltage and mixed frequency with a loss of alpha waves;	Eyelids begin to droop, the eyes may rove slowly from side to side, and the pupils become smaller
N2	0.5- to 2-s bursts of biparietal 12- to 14-Hz waves (sleep spindles) and intermittent high-amplitude, central-parietal sharp slow-wave complexes appear (vertex waves)	
N3	Appearance and increasing proportion of high-amplitude delta waves (0.75- μ V, 0.5-to 2-Hz) in the EEG	
R (REM)	Desynchronized, i.e., it has a low-voltage, high-frequency discharge pattern.	Further reduction in muscle tone except in the extraocular muscles and with bursts of rapid eye movement

Physiologic Effects of Sleep

	NREM	REM*
Autonomic Nervous System	Increased parasympathetic tone	Unstable Brief surges in sympathetic and parasympathetic activity
Cardiovascular system	Heart rate is slow and regular Blood pressure is lower	Heart rate is irregular Transient increases in blood pressure
Respiratory system	Mild decreased ventilatory drive in response to hypoxia and hypercapnia Decreased respiratory rate	Significant decrease in ventilatory drive in response to hypoxia and hypercapnia Increased respiratory rate
Temperature regulation	Decreased hypothalamic temperature set point	Decreased thermoregulation
Reproductive system		Penile erections Increased vaginal blood flow
Dreaming	Less vivid	More vivid, visually intense
Skeletal Muscle	Decreased tone with occasional movements that decrease in frequency the deeper the NREM sleep	Almost complete paralysis (REM atonia)

*REM: rapid eye movement

Sleep Disorders

Categorization of Sleep Disorders

- **Dyssomnias** – disorders of sleep-wake time and/or quality
 - Primary (intrinsic)
 - Primary insomnia
 - Sleep-related movement disorders, e.g., restless legs syndrome
 - Central disorders of hypersomnolence, e.g., narcolepsy
 - Secondary (extrinsic)
 - Secondary insomnia, e.g., depression, anxiety
 - Circadian rhythm sleep-wake disorders, e.g., jet-lag
 - Obstructive sleep apnea, congestive heart failure, GERD, acute illness, etc.
- **Parasomnias** – physical behaviors (e.g., sleepwalking) or experiences (e.g., sleep terrors) that occur during or around sleep

Insomnia

- Occurs **episodically** in up to **95%** of adults, but is **chronic** in only about **10%**
- Patients complain of not being able to fall asleep, day time sleepiness, and/or fatigue
- **Primary insomnia** (10%)
 - No secondary cause found
 - May occur after a stressful period has resolved
 - Usually resolves spontaneously

Secondary Insomnia (90%)

- Up to 50% of secondary insomnia is due to a psychiatric condition, e.g., depression, anxiety, mania, PTSD, ADHD, or OCD
- Medications: **stimulants** (caffeine, nicotine, amphetamines, cocaine), **depressants** (opiates, alcohol, barbiturates, benzodiazepines), **hallucinogens** (phencyclidine, LSD, marijuana, MDMA), **diuretics**
- Acute or chronic medical illness, e.g., COPD, SHF, BPH, acute gastroenterocolitis

PTSD = post-traumatic stress disorder, ADHD = attention deficit hyperactivity disorder, OCD = obsessive compulsive disorder, LSD = lysergic acid diethylamide, MDMA = 3,4-Methylenedioxymethamphetamine, COPD = chronic obstructive pulmonary disease, SHF = systolic heart failure, BPH = benign prostatic hypertrophy

General Treatment of Insomnia

- Treat underlying conditions or discontinue substances that promote insomnia
- Regular exercise and nutritious diet
- Sleep Hygiene
- Nonbenzodiazepine hypnotics
 - Zolpidem, Zaleplon, esZopiclone (ZZZ...)

Specific Treatment of Insomnia

- 1st Line: sleep hygiene
- 2nd: pharmacologic: ZZZ..., Melatonin receptor agonists
- 3rd: benzodiazepines* – disrupts sleep cycle, addiction
- 4th: antipsychotics, antidepressants, antihistamines – last resorts due to high side effect profile
- Above may be different for specific sleep disorders, e.g., narcolepsy

*In the past, benzodiazepines were so commonly prescribed 1st or 2nd line for insomnia that you may see them still prescribed earlier than the above scheme, a prescriber habit that should be updated.

Sleep Hygiene

- Attempt prior to medical therapy
- Wake up in the morning and go to sleep at night at same time daily
- Avoid late day caffeine, alcohol, large meals, screens (TV, computer, gaming, phone, e-Reader, etc.)
- Sleep in a dark, quiet, stress-, phone- and pet-free environment

Sleep-related movement disorders

- Restless Legs Syndrome

- Leg burning, itching, tingling, etc. that is relieved by movement
- 5%-15% of US population
- Cause unknown in most cases, but in few...
- RLS may be secondary to pregnancy, iron deficiency, peripheral neuropathy, chronic kidney disease, ADHD
- Treatment: pregabalin, dopamine agonists, gabapentin. RLS associated with pregnancy usually resolves after delivery

Circadian Rhythm Sleep-Wake Disorders

- Jet lag
 - Treatment: melatonin, stay-awake while the sun is up
- Uncommon:
 - Delayed sleep-phase disorder – going to bed and waking up late
 - Blindness, night shift work
 - Advanced sleep-phase disorder - going to bed and waking up early
 - Treatment: sleep hygiene, light therapy, melatonin receptor agonist (tasimelteon)

Central Disorders of Hypersomnolence

• Narcolepsy

- REM intrusion into wakefulness
 - Sleep paralysis - inability to talk or move on awakening or falling asleep
 - Excess hypnopompic or hypnagogic hallucinations
 - Cataplexy after strong emotions may result in trauma
- Excessive daytime sleepiness
- Onset young adults, non-progressive
- <0.2% of US population, peaks at 15 and 35 years old
 - 90% have HLA-DQB1*0602 allele (present in those without narcolepsy only 12-34%)
- Decreased orexin production, may follow viral or bacterial infections
- Diagnosis
 - Polysomnogram shows rapid sleep onset, early REM and night awakening
 - Multiple sleep latency test: shorter time to enter REM associated with more severe disease
- Treatment
 - Sleep hygiene
 - Stimulants: modafinil, methylphenidate, methamphetamine, SSRIs, serotonin and norepinephrine reuptake inhibitors (e.g., duloxetine), TCAs, sodium oxybate
- Prognosis
 - Excellent with treatment
 - Comorbid depression and anxiety

HLA = human leukocyte antigen, SSRI = selective serotonin reuptake inhibitor, TCA = tricyclic antidepressant

[C1B](#); [S1B](#)

Sleep-related Breathing Disorders

- Obstructive sleep apnea
 - Mechanical obstruction by excess soft tissues resulting in rapid, repetitive airway collapse and hypooxygenation
 - Risk: increased age, obesity, large tonsils/neck, PCOD, hypothyroidism, genetic predisposition, SHF, CKF, alcohol, smoking, male
 - Prevalence: 32% in U.S. adults > 20 years; 59% male, 41% female
 - Results in snoring and inability to reach deep sleep (often brought to medical attention by sleep partner)
 - Diagnosis: Polysomnogram
 - Treatment:
 - Continue positive airway pressure (CPAP) is treatment of choice
 - Facemask strapped to face delivers air at increased pressure to open airways during inhalation
 - Uvulopalatopharyngoplasty (UPPP)
 - Repositioning or resection of uvula and tonsils
 - Enlargement of pharynx
 - Rarely required



PCOD = polycystic ovary disease, SHF = systolic heart failure, CKF = chronic kidney failure
[Resp Med 2025](#); [C1B](#); [S1B](#)

Parasomnias (N3)

- Sleep Terror Disorder
 - Night awakening in inconsolable panic and may be associate with tachycardia, tachypnea and diaphoresis
 - No subsequent memory of the event after waking in the morning
 - Affects 1% of children
 - Usually resolves spontaneously

Parasomnias (N3)

- Sleepwalking aka somnambulism
 - Despite being physiologically asleep, the person appears awake and performs activities and have no memory of the event the next morning
 - Occurs in up to 30% of the US population as a single, non-repeating event.
 - Chronic sleepwalking may affect up to 2%
 - Usually resolves spontaneously in children
 - Treatment in adults: safety, treat underlying medical or psychological conditions, benzodiazepines

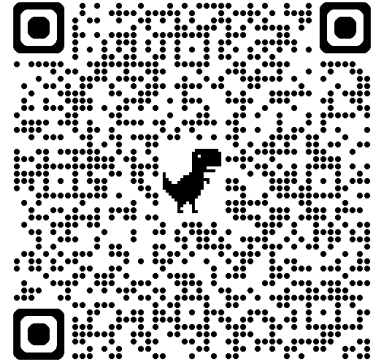
Parasomnias (REM)

- REM Sleep Behavior Disorder
 - Acting out dreams by yelling, kicking, punching, grabbing, or jumping
 - Usually older persons who remember the dreams, but not their behavior while sleeping
 - Polysomnogram shows REM sleep despite lack of atonia
 - Treatment: melatonin or low-dose clonazepam
 - Brainstem histology shows Lewy bodies that may be associated with Parkinson's, multisystem atrophy and Lewy body dementia

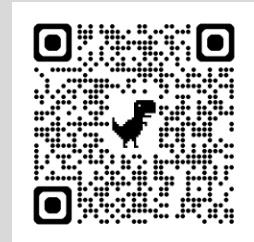
Systems-based Practice: Patient Safety

Consequences of Sleep Deprivation

- Fatal familial insomnia, a rare prion disease, results in progressive insomnia and eventual death
- A few days without sleep causes progressive slowed, altered cognition and motor control; hallucinations; paranoia; memory loss; and psychosis
- Guinness record for longest awake is 18 days, 21 hours and 40 minutes without sleep
- Guinness no longer monitors or records sleep deprivation due to the negative health effects
- The risk of motor vehicle accidents increases significantly with sleep deprivation



GuinnessWorldRecords.com



BMC Medicine

Effects of Sleep Deprivation: Impaired Cognition

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Sleepiness and Safety

Effect of Sleep Deprivation on Driving Safety in Housestaff

Carole L. Marcus and Gerald M. Loughlin



Procedia

Social and Behavioral Sciences

Effects of driver task-related fatigue on driving performance

Massimiliano Gastaldi ^{a,*}, Riccardo Rossi ^a, Gregorio Gecchele ^a



The Cumulative Cost of Additional Wakefulness: Dose-Response Effects on Neurobehavioral Functions and Sleep Physiology From Chronic Sleep Restriction and Total Sleep Deprivation

Hans P.A. Van Dongen, PhD;¹ Greg Maislin, MS, MA;¹ Janet M. Mullington, PhD;² David F. Dinges, PhD¹



[Marcus C., Loughlin G. Sleep. 1996;19:763-6](#)

[Van Dongen H., et al. Sleep. 2003;26:117-26](#)

[Gastaldi M., et al. Procedia Soc Behav Sci 2014;111:955-64](#)

Effects of Sleep Deprivation: Impaired Decision Making

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Impaired decision making following 49 h of sleep deprivation
WILLIAM D. S. KILLGORE, THOMAS J. BALKIN and NANCY J.
WESENSTEN

Journal of
Sleep
Research



One Night of Sleep Loss Impairs Innovative
Thinking and Flexible Decision Making

Y. Harrison, J.A. Horne

Sleep Deprivation Elevates Expectation of Gains and Attenuates Response to
Losses Following Risky Decisions

Vinod Venkatraman, MEng^{1,3}; YM Lisa Chuah, PhD¹; Scott A Huettel, PhD^{2,3}; Michael WL Chee, MBBS, MRCP(UK)^{1,2}

SLEEP
Official Publication of the Sleep Research Society



Caffeine Effects on Risky
Decision Making After 75
Hours of Sleep Deprivation

Authors: Killgore, William D. S.; Lipizzi, Erica L.; Kamimori, Gary H.; Balkin, Thomas J.



[Killgore, W., et al. J Sleep Res 2006;15:7-13](#)

[Harrison, Y., Horne, J. Org Behavior Hum Decision Making. 1999;78:128-45.](#)

[Venkatraman, V., et al. Sleep. 2007;30:603-9](#)

[Killgore, W., et al. Aviat Space Environ Med. 2007;78:957-62](#)

Sleep Deprivation: Infection

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Sleep Habits and Susceptibility to the Common Cold

Sheldon Cohen, PhD; William J. Doyle, PhD; Cuneyt M. Alper, MD;
Denise Janicki-Deverts, PhD; Ronald B. Turner, MD

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Official Publication of the Sleep Research Society

BEHAVIORALLY ASSESSED SLEEP AND SUSCEPTIBILITY TO THE COMMON COLD

Behaviorally Assessed Sleep and Susceptibility to the Common Cold

Aric A. Prather, PhD¹; Denise Janicki-Deverts, PhD²; Martica H. Hall, PhD³; Sheldon Cohen, PhD²

SLEEP AND ANTIBODY RESPONSE TO HEPATITIS B VACCINATION

<http://dx.doi.org/10.5665/sleep.1990>

Sleep and Antibody Response to Hepatitis B Vaccination

Aric A. Prather, PhD^{1,2}; Martica Hall, PhD³; Jacqueline M. Fury, BS¹; Diana C. Ross, MSN, RN¹; Matthew F. Muldoon, MD, MPH⁴; Sheldon Cohen, PhD²;
Anna L. Marsland, PhD, RN¹

PROSPECTIVE STUDY OF SLEEP DURATION AND PNEUMONIA RISK IN WOMEN

<http://dx.doi.org/10.5665/sleep.1594>

A Prospective Study of Sleep Duration and Pneumonia Risk in Women

Sanjay R. Patel, MD, MS¹; Atul Malhotra, MD¹; Xiang Gao, MD, PhD²; Frank B. Hu, MD, PhD²; Mark I. Neuman, MD, MPH³; Wafaie W. Fawzi, MD, DrPH^{2,4}

[Cohen, S., et al. Arch Intern Med 2009;169:62-7](#)

[Prather, A., et al. Sleep. 2015;38:133-9](#)

[Prather, A., et al. Sleep. 2012;35:1063-9](#)

[Patel, S., et al. Sleep. 2012;35:97-101](#)

Thank you for reviewing and good luck!

- Summary

1. Normal sleep physiology is dictated by circadian rhythms that orchestrate hormonal and physiologic functions that maintain normal homeostasis
2. Sleep disorders are prevalent and many can be diagnosed by a primary provider with sleep studies while some require outside consultation for management
3. Sleep is an integral part of physician wellness that optimizes safety and care of the patient

- [Feedback](#)